

UrbanLoc: Language-Guided User Localization via Location-Aware Multimodal Retrieval

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Abstract—To dispatch robots to appropriate pickup or drop-off points, autonomous personal mobility and on-demand robotic delivery services require user locations to be estimated in diverse environments. Although Global Navigation Satellite System-based user localization is widely used for this purpose, it does not always function reliably because of signal degradation in urban canyons or in situations where sensing is restricted. In contrast, language-guided user localization offers a complementary, hardware-agnostic alternative, allowing users to describe their surroundings in natural language under modality-restricted conditions. Such language-guided user localization is challenging because users typically refer only to salient landmarks or text on signs, leading to a many-to-many correspondence between linguistic descriptions and possible locations in large-scale environments. To address this, we propose UrbanLoc, a two-stage language-guided user localization method via multimodal retrieval over geotagged street-view images. Unlike existing approaches, UrbanLoc leverages complex vision-language cues (e.g., storefront signage) that users typically describe in practical large-scale environments. To this end, we introduce location-aware vision-language alignment using area summaries, along with multimodal large language model-based reranking using object coverage. To validate UrbanLoc, we constructed a benchmark consisting of 34k descriptions referring to 15k locations across 200 km² regions. Experimental results show that UrbanLoc outperforms baseline methods on standard evaluation metrics.

I. INTRODUCTION

The dispatch of mobility assistance and emergency robotic responders fundamentally depends on accurately estimating user location across diverse environmental conditions. While many user localization systems primarily rely on Global Navigation Satellite System (GNSS) [1], inertial localization [2], WiFi fingerprinting [3], or cameras [4], [5], such sensing modalities are not always available in practice. Specifically, GNSS signals can be degraded in urban canyons, and other modalities may be restricted due to connectivity limitations or privacy concerns [1]. As a result, user location estimation in modality-restricted settings remains insufficiently addressed, hindering universal autonomous mobility dispatch in real-world scenarios. In this context, language-guided user localization could offer a hardware-agnostic, non-continuous sensing alternative, as users can describe their surroundings even in constrained environments [6]–[9].

In this study, we focus on the task of predicting target locations given user-provided descriptions of their surroundings. Fig. 1 shows a typical scenario for this task. The user is located in an urban canyon, where GNSS accuracy is often degraded. The user requests a pickup with a smartphone by describing the surrounding environment (e.g., “I am standing under the traffic light at the intersection lined with tall brown



Fig. 1: **Typical scenario for this task.** The user requests a pickup from a location where sensing modalities are restricted by describing the surrounding environment using a smartphone. UrbanLoc predicts the user’s location, and a robot is dispatched to the estimated destination.

buildings, right in front of the white shop with the ‘LOS TACOS’ sign”). UrbanLoc predicts the user’s location, and a robot is dispatched to the estimated destination. Once the destination coordinates have been specified, the robot can rely on a high-precision map for navigation; therefore, the primary challenge lies in estimating the user’s location.

This task is challenging because users typically refer only to salient landmarks and/or text on signs, leading to a many-to-many correspondence between linguistic descriptions and possible locations in large-scale environments. Indeed, multimodal large language models (MLLMs) [10], [11] often fail to accurately predict the target location, as a given description can yield semantically plausible but geographically unrelated candidates. Furthermore, conventional methods sometimes rely on point clouds [7], [8] and satellite images [9], which often lack complex vision-language cues described by the user (e.g., storefront signage and spatial layout). Moreover, they do not explicitly incorporate area-level cues (e.g., features shared across a street), which are often crucial for user localization.

To address these limitations, we propose UrbanLoc, a two-stage language-guided user localization method via multimodal retrieval and MLLM-based reranking over geotagged street-view images. We introduce location-aware vision-language alignment with area summaries to mitigate the partial and viewpoint-dependent nature of a single location observation. We also introduce MLLM-based reranking using object coverage because vision-language similarity alone

may over-rank locations that are semantically plausible yet geographically unrelated.

The main contributions of this study are as follows:

- We propose UrbanLoc, a two-stage language-guided user localization method via multimodal retrieval over geotagged street-view images.
- We introduce location-aware vision-language alignment using area summaries, along with MLLM-based reranking using object coverage.
- We construct the UrbanLoc-Text benchmark, which consists of 34k descriptions referring to 15k locations across 200 km² regions.

II. RELATED WORK

A. Modality-Dependent User Localization

Unlike robot localization, user localization typically assumes that only commodity user devices (e.g., smartphones) are available and that robot-oriented sensors (e.g., wheel encoders) are unavailable [12], [13]. Therefore, many standard robot localization methods based on robot-oriented sensors have limited applicability to user localization.

Previous user localization studies often use modalities such as GNSS [1], [2], inertial measurement units (IMUs) [2], WiFi fingerprints [3], or cameras [4], [5]. For example, Yi et al. [2] integrate GNSS and IMU measurements with the Galileo High Accuracy Service for outdoor user localization, and DeepFi [3] models WiFi channel state information fingerprints for indoor user localization. Moreover, image-based user localization methods have been studied. For example, classification-based methods (e.g., [4]) classify queries into geographic cells, whereas cross-view retrieval-based methods (e.g., [5]) match ground-level queries with aerial images. Recent visual geo-localization methods, including cross-view matching and object-level geo-localization are extensively reviewed in [14], [15]. However, such modalities are not always available or reliable because of environmental factors, connectivity limitations, or privacy concerns. Specifically, GNSS can be degraded in urban canyons due to building blockage and multipath conditions [1].

B. Language-Guided Localization

Language-guided user localization could offer a hardware-agnostic, non-continuous sensing alternative, as users can describe their surroundings even in constrained environments [6]–[9]. Existing studies span different platforms and map representations. For aerial platforms, CityNav [6] identifies language-specified destinations in photorealistic 3D city environments using drone observations. Similarly, Text2Loc [7] and CMMLoc [8] associate natural language descriptions with outdoor point-cloud maps for autonomous driving to estimate the described location. Furthermore, CrossText2Loc [9] estimates a geographic coordinate from ground-level descriptions by retrieving geotagged satellite images or OpenStreetMap (OSM) renderings. Related street view studies [16]–[18] explore embodied navigation and instruction following in real urban visual environments, with a primary emphasis on route execution and local spatial reasoning in street-view graphs.

C. Multimodal Retrieval Using Foundation Models

Language-guided localization with multimodal foundation models has been widely studied [19]–[26]. For example, multimodal retrieval over pre-explored embodied memories has been used to determine target objects or places given natural language instructions [27], [28]. In these pipelines, vision-language models such as CLIP [19] and SigLIP [20] are often used as retrieval backbones. Other multimodal retrieval studies are comprehensively reviewed in [29], [30].

Following the CLIP-style contrastive formulation, GeoCLIP [31] aligns embeddings of GNSS coordinates with image embeddings in a shared embedding space. GT-Loc [32] further incorporates time into the embeddings, enabling the estimation of both where and when an image was captured. Beyond image-based approaches, language-guided user localization can be formulated as text-image retrieval over map views [9]. CrossText2Loc [9] estimates the user location from free-form text by retrieving satellite images or OSM renderings, which often lack complex vision-language cues and thus may limit user localization performance.

D. Benchmarks for Language-Guided Localization

Several benchmarks have been introduced for language-guided localization. For example, GeoText-1652 [33] is a representative benchmark for drone localization that provides images captured by unmanned aerial vehicles, together with spatial descriptions and phrase-level correspondences. By contrast, robot localization datasets such as KITTI [34] are primarily designed for robot localization and are not directly applicable to language-guided user localization. However, their driving scenes can be repurposed for this setting, as exemplified by KITTI360Pose [35], which is built on KITTI-360 [36]. Most relevant to our setting, CVG-Text [9] is a benchmark for language-guided user localization, where ground-level textual queries are associated with satellite images or OSM-rendered views.

III. PROBLEM STATEMENT

In this study, we focus on the task of predicting target locations given user-provided descriptions of their surroundings. In this task, the model is expected to estimate the geographic coordinates corresponding to a given description. Fig. 1 shows a typical scenario for this task.

We assume that the user is located in a densely built urban area, where meter-level location estimates are unavailable. This setting is reasonable because GNSS can be unreliable in dense urban environments due to non-line-of-sight and multipath effects, where positioning errors can exceed 100 m [1]. We also assume that a service provider has access to a collection of geotagged images because survey-grade geotagged images are widely accessible through public APIs provided by major mapping platforms such as Mapillary (<https://www.mapillary.com/>).

IV. PROPOSED METHOD

We propose UrbanLoc, a two-stage language-guided user localization method via multimodal retrieval over geotagged street-view images. We formulate this task as text-to-image retrieval because a text-only localizer would struggle to

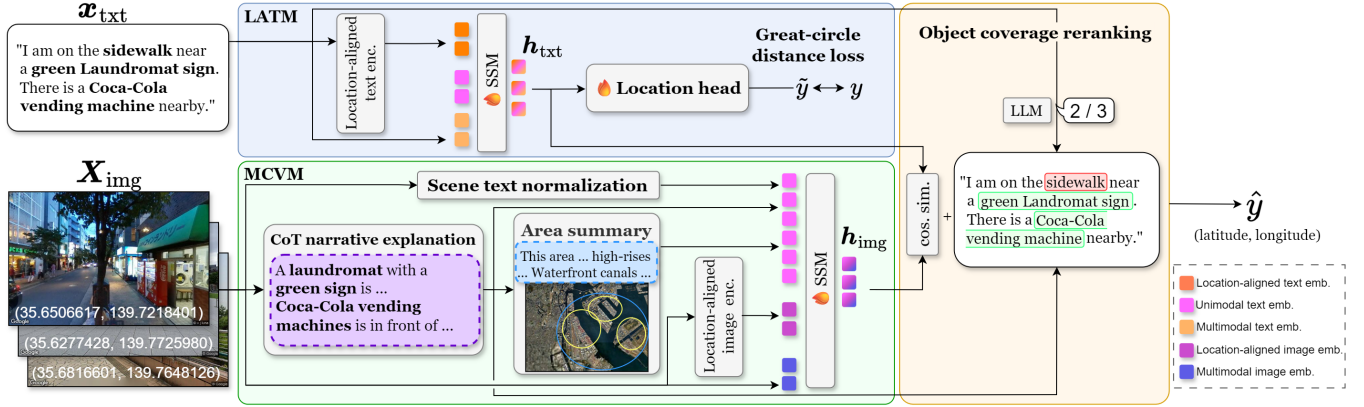


Fig. 2: **Architecture of UrbanLoc.** LATM obtains location-aware *text* representations by introducing an auxiliary location head. Location-aware *visual* representations are obtained in MCVM by aggregating cues shared across a street. Finally, UrbanLoc reranks candidate locations based on object coverage. $\tilde{y}$, $\hat{y}$, and y denote the location predicted by the auxiliary location head, the final estimated location, and the ground-truth location, respectively. LATM, MCVM, and SSM represent Location-Aware Text Module, Multi-Cue Visual Module, and a deep state space model encoder, respectively.

associate appearance-dependent cues with geographic coordinates. For example, in a description such as “There is a red and white signboard that says ‘Joe’s Pizza’ at the entrance of a brown brick building,” the cues are more reliably utilized in a text-to-image retrieval setting, where they can be grounded against candidate images in X_{img} .

Fig. 2 shows the architecture of UrbanLoc. Our model consists of two modules, Location-Aware Text Module (LATM) and Multi-Cue Visual Module (MCVM), together with object coverage reranking. LATM obtains a location-aware text representation using an auxiliary location head with a great-circle distance loss to reflect geographic proximity. MCVM includes (i) a chain-of-thought (CoT) narrative representation that captures multiple objects and their fine-grained attributes and (ii) an area-level summary that captures characteristics shared across nearby locations. UrbanLoc reranks candidate locations based on object coverage. Each module can be applied to other tasks where location information plays an important role, such as vision-language navigation and street-scene or map-based visual question answering [37], [38].

The input to the model consists of x_{txt} and $X_{\text{img}} = \{x_{\text{img}}^{(i)}\}_{i=1}^{N_{\text{img}}}$, where x_{txt} denotes a description and $x_{\text{img}}^{(i)} \in \mathbb{R}^{3 \times W \times H}$ denotes a geotagged image. When indexing multiple descriptions, we denote the j -th description by $x_{\text{txt}}^{(j)}$. Here, i , N_{img} , W , and H denote the index of each image, the number of candidate images for retrieval, and the width and height of each image, respectively.

A. Location-Aware Text Module

LATM maps x_{txt} to a location-aware text representation h_{txt} . In our task, x_{txt} can match multiple places. As a result, existing vision-language embedding models (e.g., [19], [20], [24]) may struggle to rank candidate locations reliably, because their image-text similarity alone may not sufficiently distinguish the target location from other plausible candidates. This motivates the incorporation of geographic structure into the representations. Although GeoCLIP [31] aligns visual representations with geographic coordinates, its supervision is not given in terms of geographic proximity, which is central to our task. To address this, we introduce an

auxiliary location head that estimates geographic coordinates from x_{txt} , supervised by a great-circle distance regression loss (see Eq. (6)).

Given x_{txt} , we obtain h_{txt} as follows. First, we apply location-aligned, unimodal, and multimodal text encoders to x_{txt} and concatenate the resulting features. Then, the concatenated feature vector is input to a deep SSM [39] to obtain h_{txt} . The auxiliary location head predicts the geographic coordinates $\tilde{y}$.

B. Multi-Cue Visual Module

To capture both visual and location-related cues, MCVM combines visual appearance features with geographic features. This module includes the following three main submodules.

1) **CoT Narrative Representation:** Because visual representations alone can miss object-level and relational semantics, we generate a natural-language explanation of $x_{\text{img}}^{(i)}$ and embed it as a narrative representation [40]. To mitigate hallucinations, we generate the explanation via a two-phase CoT-style reasoning pipeline. In the first phase, an MLLM generates a structured list of distinctive objects and their attributes from $x_{\text{img}}^{(i)}$. This list helps ground the final explanation to visually observed cues. In the second phase, given this list, the MLLM generates a CoT narrative explanation of $x_{\text{img}}^{(i)}$ (visualized as the purple text block in Fig. 2). Finally, we apply a unimodal text encoder to the explanation to obtain the CoT narrative representation.

2) **Area Summary:** This submodule summarizes a local area to leverage the characteristics shared across nearby locations. Because $x_{\text{img}}^{(i)}$ provides only a partial and viewpoint-dependent observation, it is insufficient to capture the area-level cues useful for user localization (e.g., architectural styles and street network patterns). Although methods for aggregating area-level information (e.g., [27]) can mitigate this limitation, naively aggregating all location explanations within each area is computationally infeasible when the area contains many locations.

Instead, we obtain an area summary feature as follows. We compute an area mean vector from the CoT narrative

representations associated with locations within a predefined area and select representative locations closest to this mean vector. Then, an LLM generates an area summary (indicated by the blue text block in Fig. 2) from the CoT narrative explanations of the selected locations. Finally, we apply a unimodal text encoder to the summary to obtain the area summary feature.

3) **Scene Text Normalization:** To mitigate a language mismatch between $\mathbf{x}_{\text{txt}}$ and multilingual optical character recognition (OCR) text, we perform language normalization by translating non-named-entity tokens and converting named entities into brief explanations. In many parts of the world, street scenes contain multilingual text (e.g., shop signs and notices). As a result, directly embedding OCR text would be ineffective if there is a language mismatch with $\mathbf{x}_{\text{txt}}$, and naive translation often fails to preserve the semantics of named entities that lack direct equivalents.

Therefore, we construct the normalized scene text feature as follows. We provide the scene text detected in $\mathbf{x}_{\text{img}}^{(i)}$ to a multilingual MLLM (e.g., [41]), which normalizes it into the language of $\mathbf{x}_{\text{txt}}$ by generating brief explanations for named entities and translating the remaining text (e.g., explaining “El Corte Inglés” as “a department store”). We then embed the normalized text using a unimodal text encoder.

Given $\mathbf{x}_{\text{img}}^{(i)}$, we obtain the final output of MCVm ($\mathbf{h}_{\text{img}}^{(i)}$) as follows. First, we encode $\mathbf{x}_{\text{img}}^{(i)}$ using both a location-aligned image encoder [31] and a multimodal image encoder [24]. We then concatenate the output features from the two image encoders with the CoT narrative representation, the area summary feature, and the normalized scene text feature. Finally, the concatenated feature vector is processed by a deep SSM, yielding $\mathbf{h}_{\text{img}}^{(i)}$. Because $\mathbf{h}_{\text{img}}^{(i)}$ can be precomputed as part of our offline map construction pipeline, it adds no image-encoding cost at inference time.

C. Object Coverage Reranking

To enable object-level agreement between $\mathbf{x}_{\text{txt}}^{(j)}$ and $\mathbf{x}_{\text{img}}^{(i)}$, UrbanLoc reranks $\mathbf{X}_{\text{img}}$ based on an object coverage rate between $\mathbf{x}_{\text{txt}}^{(j)}$ and the CoT narrative explanation of $\mathbf{x}_{\text{img}}^{(i)}$. A single global cosine similarity score may be overly influenced by a salient object that matches only some of the objects specified in $\mathbf{x}_{\text{txt}}^{(j)}$. As a result, when $\mathbf{x}_{\text{txt}}^{(j)}$ refers to multiple objects, as exemplified in Fig. 2, partial-match candidates may be ranked above those that match all objects specified in $\mathbf{x}_{\text{txt}}^{(j)}$. To address this, we introduce a reranking term based on the object coverage rate.

During inference, UrbanLoc first computes the cosine similarity between $\mathbf{h}_{\text{txt}}^{(j)}$ and $\mathbf{h}_{\text{img}}^{(i)}$ for each pair $(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)})$:

$$s(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)}) = \frac{\mathbf{h}_{\text{txt}}^{(j)} \cdot \mathbf{h}_{\text{img}}^{(i)}}{\|\mathbf{h}_{\text{txt}}^{(j)}\| \|\mathbf{h}_{\text{img}}^{(i)}\|}. \quad (1)$$

UrbanLoc then reranks $\mathbf{X}_{\text{img}}$ using the following score $s'(\cdot, \cdot)$:

$$s'(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)}) = s(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)}) + \lambda_{\text{obj}} R_{\text{obj}}^{(i,j)}, \quad (2)$$

TABLE I: Statistics of the UrbanLoc-Text benchmark.

	Train/Val		Test	
	New York	Tokyo	New York	Tokyo
Descriptions	12,298	14,790	3,197	3,643
Locations	6,149	7,395	647	763
Images	6,149	7,395	647	763
Vocabulary size	10,254	11,323	3,823	4,613
Word count	515,234	637,535	88,196	101,112
Average length	41.90	43.11	27.59	27.76
Annotators	—		151	

TABLE II: Experimental settings of UrbanLoc.

Epochs	50
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.98$)
Batch size	256
Learning rate	1×10^{-5}
LR schedule	Cosine annealing
Warmup epochs	5
λ_{obj} , λ_{dist}	0.5, 0.2
k	20

where λ_{obj} and $R_{\text{obj}}^{(i,j)} \in [0, 1]$ denote a weighting hyperparameter and the object coverage rate, respectively. $R_{\text{obj}}^{(i,j)}$ is calculated as the fraction of objects in $\mathbf{x}_{\text{txt}}^{(j)}$ that are matched by objects described in the CoT narrative explanation of $\mathbf{x}_{\text{img}}^{(i)}$. In our prompt, we provide a definition of an object and instruct an LLM to decompose both $\mathbf{x}_{\text{txt}}^{(j)}$ and the CoT narrative explanation into object sets, match objects between the two sets, and output $R_{\text{obj}}^{(i,j)}$. Here, each object is defined as an entity together with its specified attributes. For example, $\mathbf{x}_{\text{txt}}^{(j)}$ in Fig. 2 specifies three objects: the sidewalk, a green laundromat sign, and a Coca-Cola vending machine. When only two of these three objects are matched by the objects described in the CoT narrative explanation, $R_{\text{obj}}^{(i,j)} = 0.67$. For computational efficiency, we compute $R_{\text{obj}}^{(i,j)}$ only for the top- k candidates ranked by $s(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)})$ and set $R_{\text{obj}}^{(i,j)} = 0$ for the remaining candidates.

The model outputs $\hat{\mathbf{y}} = \mathbf{y}^{(i^*)}$, where $i^* = \arg \max_i s'(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)})$. In addition, $\mathbf{y}^{(i)}$ denotes the geographic coordinates of $\mathbf{x}_{\text{img}}^{(i)}$.

We use the following loss:

$$\mathcal{L} = \mathcal{L}_{\text{t2i}} + \mathcal{L}_{\text{i2t}} + \lambda_{\text{dist}} \mathcal{L}_{\text{dist}}, \quad (3)$$

$$\mathcal{L}_{\text{t2i}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(s(\mathbf{x}_{\text{txt}}^{(i)}, \mathbf{x}_{\text{img}}^{(i)})/\tau)}{\sum_{j=1}^B \exp(s(\mathbf{x}_{\text{txt}}^{(i)}, \mathbf{x}_{\text{img}}^{(j)})/\tau)}, \quad (4)$$

$$\mathcal{L}_{\text{i2t}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(s(\mathbf{x}_{\text{txt}}^{(i)}, \mathbf{x}_{\text{img}}^{(i)})/\tau)}{\sum_{j=1}^B \exp(s(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)})/\tau)}, \quad (5)$$

$$\mathcal{L}_{\text{dist}} = \frac{1}{B} \sum_{i=1}^B d_{\text{gc}}(\tilde{\mathbf{y}}^{(i)}, \mathbf{y}^{(i)}), \quad (6)$$

where $d_{\text{gc}}(\cdot, \cdot)$ and $\tilde{\mathbf{y}}^{(i)}$ denote the great-circle distance using the Haversine formula [42] and the output of the auxiliary location head for the i -th description, respectively. In addition, λ_{dist} , B , and τ denote a weighting hyperparameter, a batch size, and a temperature parameter, respectively.

TABLE III: **Quantitative comparison** of UrbanLoc and the baseline methods on the New York and Tokyo subsets of the UrbanLoc-Text benchmark. Here, “FT” indicates whether the model was fine-tuned. For these methods, the mean and standard deviation over five trials are reported. The highest score for each metric is in **bold**, and the second-highest score is underlined.

[%] Method	FT	New York			Tokyo		
		L@50↑	R@1↑	R@5↑	L@50↑	R@1↑	R@5↑
CLIP [19]	❄	10.5	10.1	19.1	11.1	10.8	22.0
CLIP	🔥	18.9 ±0.1	18.3 ±0.1	31.8 ±0.1	19.3 ±0.2	18.8 ±0.2	34.2 ±0.1
BEiT-3 [23]	❄	14.6	14.1	29.0	14.9	14.5	31.9
BEiT-3	🔥	17.5 ±0.2	16.6 ±0.2	31.6 ±0.4	17.8 ±0.2	17.3 ±0.3	34.7 ±0.4
BLIP-2 [22]	❄	6.4	6.2	14.1	7.7	7.2	16.8
BLIP-2	🔥	17.9 ±0.0	17.7 ±0.0	34.2 ±0.1	19.8 ±0.0	19.4 ±0.0	37.6 ±0.1
SigLIP [20]	❄	16.1	15.5	28.3	16.4	16.2	30.6
SigLIP	🔥	21.5 ±0.2	20.7 ±0.2	35.3 ±0.2	18.3 ±0.2	17.8 ±0.2	34.5 ±0.1
Long-CLIP [25]	❄	11.4	10.8	21.7	13.1	12.7	26.2
Long-CLIP	🔥	17.2 ±0.1	16.6 ±0.1	30.9 ±0.1	18.4 ±0.1	17.8 ±0.1	34.5 ±0.1
CrossText2Loc [9]	❄	1.2	1.0	3.1	1.1	0.9	2.6
CrossText2Loc	🔥	2.5 ±0.1	2.2 ±0.0	5.8 ±0.0	3.2 ±0.1	2.9 ±0.0	4.5 ±0.1
SigLIP 2 [21]	❄	15.1	14.9	28.4	14.5	14.2	29.6
SigLIP 2	🔥	19.7 ±0.1	18.9 ±0.0	34.3 ±0.0	22.7 ±0.1	22.1 ±0.1	40.7 ±0.1
Meta CLIP 2 [26]	❄	13.5	13.0	23.8	15.1	14.8	27.8
Meta CLIP 2	🔥	15.6 ±0.1	15.2 ±0.1	27.8 ±0.1	18.3 ±0.1	17.8 ±0.1	32.8 ±0.1
Qwen3-VL-Embedding [24]	❄	17.3	16.7	30.0	18.8	18.3	31.8
Qwen3-VL-Embedding	🔥	17.9 ±0.6	17.2 ±0.5	31.2 ±1.0	19.2 ±0.5	18.8 ±0.6	32.9 ±0.8
Qwen3-VL-Embedding + Qwen3-VL-Reranker	🔥	<u>26.8</u> ±0.3	<u>25.9</u> ±0.4	<u>36.3</u> ±0.3	<u>26.6</u> ±0.3	<u>26.0</u> ±0.3	<u>36.5</u> ±0.4
UrbanLoc (ours)	🔥	30.2 ±0.2	28.9 ±0.2	44.9 ±0.5	31.8 ±0.4	31.0 ±0.3	49.4 ±0.7

V. EXPERIMENTS

A. Experimental Setup

Benchmark. To validate our method, we constructed the UrbanLoc-Text benchmark for the task of predicting a target location from a description. Under the assumption stated in Section III, a suitable benchmark for this task should include geotagged street-view images that can be used to train and/or test models. However, benchmarks that satisfy these two requirements remain limited. Indeed, CVG-Text [9], a representative related benchmark, lacks geotagged street-view images and contains satellite images derived from Google Maps content; Google’s terms of use prohibit the use of such content even for testing models.

Therefore, we collected geotagged street-view images for the UrbanLoc-Text benchmark via Mapillary in accordance with Mapillary’s terms of use, using predefined geographic coordinates as sampling locations. Next, we filtered the images to exclude those dominated by walls, cars, or people, as they do not provide landmarks that a user would typically refer to when describing their surroundings. Each remaining image was then paired with one or more descriptions.

We obtained descriptions for the test set and for the training/validation sets using the following two procedures. (1) Test-set descriptions were collected from human annotators. They were presented with street-view images, OCR-detected text, and annotation instructions. They were asked to imagine standing at the depicted location and describe it as if they were conveying the scene to another person via a phone call or a messaging app. Data collection was facilitated by SoSci Survey (<https://www.sosicisurvey.de/>). The collected descriptions were translated into English using

Claude [43]; however, this model was not used in our method or for training/validation description generation. To enable reliable evaluation, we manually checked every test sample after annotation according to our test-set quality-check criteria.

(2) Descriptions for the training and validation sets were generated using Qwen3-VL [11] given street-view images, OCR-detected text, and annotation instructions.

Table I presents the statistics of the UrbanLoc-Text benchmark. Each annotator provided 45.3 descriptions on average. The benchmark was geographically divided into training, validation, and test sets to reduce spatial overlap between splits. Specifically, every pair of locations from different splits was separated by at least 20 m. The training, validation, and test sets contained 24,194, 2,894, and 6,840 samples, respectively. We used the training set to train the model, the validation set to tune the hyperparameters, and the test set to evaluate model performance.

Implementation Details. Table II shows the experimental settings of the proposed method. Our model has approximately 32.7M trainable parameters and 4.28G multiply-add operations. We trained our model on a GeForce RTX 4090 with 24 GB of VRAM. The training time for the proposed method was approximately 50 minutes. At inference time, computing $s(\mathbf{x}_{\text{txt}}, \mathbf{x}_{\text{img}}^{(i)})$ for 647 candidate images in the New York subset took approximately 270 ms. For the test-set evaluation, we used the checkpoint with the highest recall@1 on the validation set.

Baselines. We used CLIP (ViT-L/14) [19], BEiT-3 (large) [23], BLIP-2 (ViT-g) [22], SigLIP (ViT-So/14) [20],

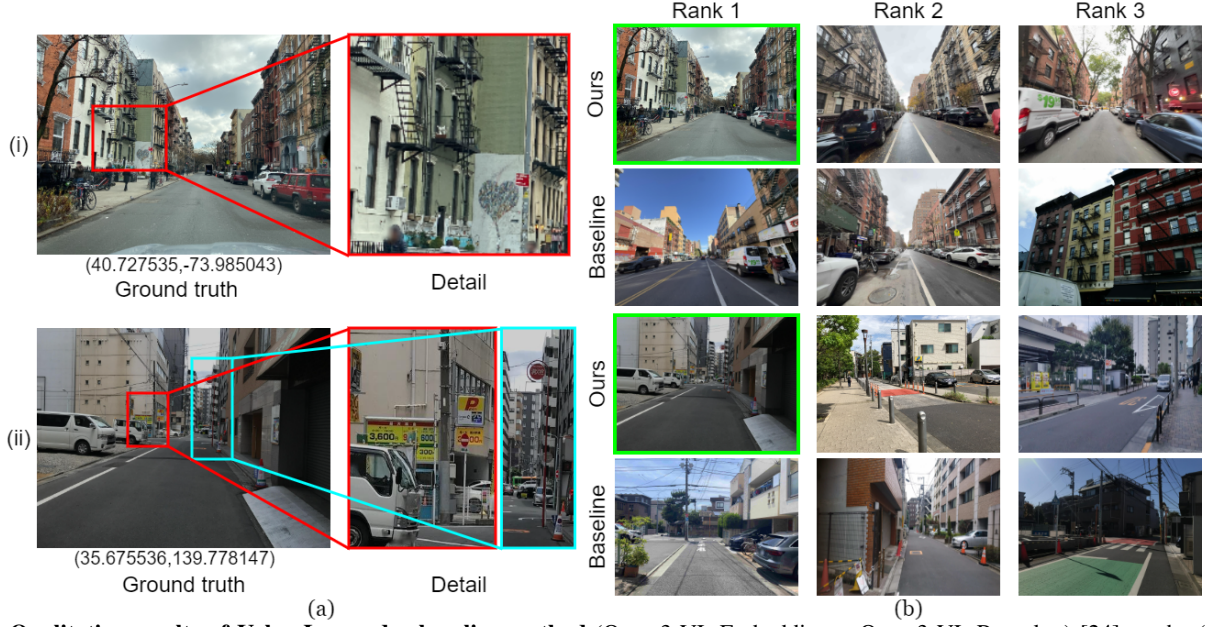


Fig. 3: **Qualitative results of UrbanLoc and a baseline method** (Qwen3-VL-Embedding + Qwen3-VL-Reranker) [24] on the (i) New York and (ii) Tokyo subsets. The corresponding x_{txt} were (i) “I am on a street with many brick buildings that have external stairs, and I can see a white sign with a black heart on the external wall to my front left.” and (ii) “There is a parking lot on the left with a yellow sign saying max 3,600 yen, and a red sign for a fire hydrant on the right.” The green frames indicate the images of the ground-truth locations.

Long-CLIP (ViT-L/14) [25], SigLIP 2 (ViT-So/14) [21], Meta CLIP 2 [26], Qwen3-VL-Embedding-2B [24], and Qwen3-VL-Embedding-2B + Qwen3-VL-Reranker-2B as baseline methods. We selected these vision-language foundation models as baselines because they have demonstrated strong performance on multimodal retrieval tasks. In addition, we used CrossText2Loc [9] because it is a representative method for multimodal retrieval-based user localization. For CrossText2Loc, we restricted the retrieval targets to OSM images because of terms-of-use concerns.

Metrics. We used localization $\text{recall}@N$ ($L@N$, $N = 50$) [7]–[9] and $\text{recall}@K$ ($R@K$, $K = 1, 5$) as evaluation metrics, with $L@50$ as the primary metric. We used these metrics because they are standard for multimodal retrieval-based location prediction tasks [7]–[9] and multimodal retrieval [29], respectively. $L@N$ is defined as follows:

$$L@N = \frac{1}{N_{\text{txt}}} \sum_{i=1}^{N_{\text{txt}}} \mathbf{1} \left[d_{\text{gc}}(\hat{y}^{(i)}, y^{(i)}) \leq \frac{N}{1,000} \right], \quad (7)$$

where N_{txt} and $\mathbf{1}[\cdot]$ denote the number of descriptions and the indicator function, respectively.

B. Quantitative Results

Table III presents the quantitative comparison between UrbanLoc and the baseline methods on the New York and Tokyo subsets of the UrbanLoc-Text benchmark. For fine-tuned methods, we report the mean and standard deviation over five trials.

As listed in Table III, UrbanLoc achieved the best $L@50$ scores of 30.2% and 31.8% on the New York and Tokyo subsets, respectively, together with $R@1$ scores of 28.9% and 31.0%. In comparison, the strongest baseline obtained $L@50$ scores of 26.8% and 26.6%, and $R@1$ scores of 25.9% and 26.0%. The performance differences between UrbanLoc and the baseline methods were statistically significant ($p < 0.01$).



Fig. 4: **Qualitative results for a representative failure case.** The given x_{txt} was “I am beside a brown building, and across from it I can see a building that is white on the first floor and beige on the second floor and above.”

C. Qualitative Results

Fig. 3 shows the qualitative results of UrbanLoc and a baseline method (Qwen3-VL-Embedding + Qwen3-VL-Reranker) [24]. Fig. 3(i) and (ii) present samples from the New York and Tokyo subsets, respectively. In each sample, Fig. 3(a) shows $x_{\text{img}}^{(i)}$, y , and a magnified detail view, and Fig. 3(b) shows the top-3 predicted locations for UrbanLoc and the baseline method, arranged in separate rows.

Fig. 3(i) presents a case in which x_{txt} was “I am on a street with many brick buildings that have external stairs, and I can see a white sign with a black heart on the external wall to my front left.” UrbanLoc correctly ranked the target location at rank 1 by capturing building attributes and spatial relationships such as brick buildings with external stairs and a white sign with a black heart. In contrast, the baseline method ranked the target location at rank 104, with $d_{\text{gc}}(\hat{y}, y) = 1.6$ km.

In Fig. 3(ii), x_{txt} was “There is a parking lot on the left with a yellow sign saying max 3,600 yen, and a red sign for a fire hydrant on the right.” The baseline method ranked the target location at rank 85, with $d_{\text{gc}}(\hat{y}, y) = 11.9$ km, and none of its top-3 predicted locations contained any parking-lot signs. In contrast, UrbanLoc successfully ranked the target location at rank 1 by leveraging the “3,600 yen” text

TABLE IV: **Quantitative results of the ablation studies.** The mean and standard deviation over five trials are reported. The best value for each metric is in **bold**.

[%]		New York			Tokyo		
Model		L@50↑	R@1↑	R@5↑	L@50↑	R@1↑	R@5↑
(i)	UrbanLoc (full)	30.2 ± 0.2	28.9 ± 0.2	44.9 ± 0.5	31.8 ± 0.4	31.0 ± 0.3	49.4 ± 0.7
(ii)	w/o location head	28.2 ± 0.3	27.0 ± 0.4	39.8 ± 0.4	28.8 ± 0.1	28.2 ± 0.1	42.0 ± 0.3
(iii)	w/o CoT narrative representation	27.7 ± 0.4	26.4 ± 0.4	39.3 ± 0.7	28.2 ± 0.4	27.6 ± 0.4	40.7 ± 0.3
(iv)	w/o area summary	28.1 ± 0.4	26.9 ± 0.3	40.1 ± 0.2	29.4 ± 0.4	28.8 ± 0.4	42.6 ± 0.4
(v)	w/o scene text normalization	27.6 ± 0.4	26.5 ± 0.3	39.5 ± 0.4	28.9 ± 0.1	28.2 ± 0.1	42.1 ± 0.4
(vi)	w/o object coverage reranking	24.7 ± 0.3	23.6 ± 0.3	40.4 ± 0.5	25.1 ± 0.7	24.4 ± 0.7	43.1 ± 0.7

on the sign. Moreover, each of the top-3 locations predicted by UrbanLoc contained both a parking lot and its sign. These results indicate that UrbanLoc can identify the target location by taking into account the attributes and spatial relationships of multiple objects, as well as textual cues.

Fig. 4 shows a qualitative example of a failure case. Here, $\mathbf{x}_{\text{img}}^{(i)}$ and $\mathbf{y}$ are shown in the left-hand subfigure, and the top-3 locations predicted by UrbanLoc and Qwen3-VL-Embedding + Qwen3-VL-Reranker are shown in the right-hand subfigure. In this example, $\mathbf{x}_{\text{txt}}$ was “I am beside a brown building, and across from it I can see a building that is white on the first floor and beige on the second floor and above.” UrbanLoc and the baseline method ranked the target location at ranks 240 and 518, respectively. The corresponding $d_{\text{gc}}(\hat{\mathbf{y}}, \mathbf{y})$ values for UrbanLoc and the baseline method were 56.7 km and 58.0 km, respectively. This failure likely stems from the model’s difficulty in capturing fine-grained combinations of visual attributes. Specifically, each of the top-3 predicted locations contained some of the color cues described in $\mathbf{x}_{\text{txt}}$, such as brown, white, or beige building facades. However, none of these locations matched the target location with respect to the specific combination and arrangement of brown, white, and beige facade colors.

D. Ablation Study

Table IV presents the quantitative results of ablation studies. The mean and standard deviation over five trials are reported.

Location head ablation: To investigate the effect of the location head, we removed it from LATM. The L@50 scores of Model (ii) were 2.0 and 3.0 points lower than those of Model (i) on the New York and Tokyo subsets, respectively. These results suggest that incorporating location awareness into text representations through the location head is beneficial for user localization.

CoT narrative representation ablation: To assess the impact of the CoT narrative representation, we removed it to obtain Model (iii). Model (iii) had L@50 scores of 27.7% and 28.2% on the New York and Tokyo subsets, which were 2.5 and 3.6 points lower than those of Model (i), respectively. These results indicate that the CoT narrative representation enriches the visual representation by capturing fine-grained attributes and spatial relationships in each image, thereby improving vision-language alignment.

Area summary ablation: We evaluated the effectiveness of the area summary by omitting it. Table IV shows that

TABLE V: **Categorization of failure cases.**

Error category	#Errors
Compositionality error	43
Spatial comprehension error	20
Scene text comprehension error	18
Ambiguous description	12
OCR error	7
Total	100

removing this submodule (Model (iv)) decreased L@50 scores by 2.1 and 2.4 points on the New York and Tokyo subsets, respectively, compared with the full model (Model (i)). This indicates that the area summary contributes to user localization by adding coarse area-level cues.

Scene text normalization ablation: We omitted the scene text normalization submodule to analyze its contribution. Compared with Model (i), Model (v) had L@50 scores that were 2.6 and 2.9 points lower on the New York and Tokyo subsets, respectively. This suggests that scene text normalization enables our model to identify the target location based on multilingual text cues in images.

Object coverage reranking ablation: We assessed the impact of the object coverage reranking by comparing Model (i) with Model (vi), in which $\lambda_{\text{obj}} R_{\text{obj}}^{(i,j)}$ is removed from $s'(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)})$. Without the object coverage reranking, L@50 scores decreased to 24.7% and 25.1% on the New York and Tokyo subsets, respectively, corresponding to decreases of 5.5 and 6.7 points from the full model. This indicates that object-level agreement is important for improving user localization performance.

E. Error Analysis

We defined failure cases as samples with a distance greater than 50 m between $\hat{\mathbf{y}}$ and $\mathbf{y}$. There were 2,232 and 2,485 failure cases in the New York and Tokyo subsets, respectively. To investigate the limitations of our method, we analyzed 100 failure cases (50 per subset) with the largest great-circle distance errors, and the results are reported in Table V. We identified five major failure modes:

Compositionality error: This category covers cases in which the model failed to capture compositional relationships among multiple concepts, including object co-occurrence and attribute–object binding.

Spatial comprehension error: This category includes cases where the model failed to correctly understand the spatial

relationships among multiple objects in $\mathbf{x}_{\text{txt}}$, such as their relative positions or ordering.

Scene text comprehension error: This refers to cases where the model failed to properly interpret or utilize scene text information.

Ambiguous description: This covers cases in which multiple candidate locations could match the given description, making it unclear which location the description refers to.

OCR error: This involves cases where the OCR system in our pipeline failed to detect scene text in $\mathbf{x}_{\text{img}}^{(i)}$.

VI. CONCLUSIONS

In this study, we focused on the task of predicting target locations given user-provided descriptions of their surroundings. We proposed UrbanLoc, a two-stage language-guided user localization method via multimodal retrieval over geotagged street-view images. We introduced location-aware vision-language alignment using area summaries, along with MLLM-based reranking using object coverage. Furthermore, we constructed the UrbanLoc-Text benchmark, which consists of 34k descriptions referring to 15k locations across 200 km² regions in multiple cities. Experimental results show that UrbanLoc outperformed the baseline methods on standard evaluation metrics. In future work, we will incorporate an ℓ_1 -product metric over separate embedding factors to address the compositionality errors, following prior work showing that this metric can capture cross-family composition in a Boolean-algebra-like manner [44].

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